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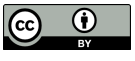
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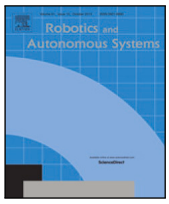
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NMPC-based variable admittance control with passivity constraints

Thibault Poignonec, Bernard Bayle, Loïc Cuvillon, Hassan Omran *

Université de Strasbourg, CNRS, INSERM, ICube, UMR7357, Strasbourg, France

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ABSTRACT

When robots interact with their environment, managing the physical interaction safely and accurately is critical. Impedance and admittance control allow such interaction and have been fundamental to robotic control for decades. For challenging applications, it is often necessary to implement a time-varying impedance profile, which necessitates more complex conditions to guarantee stability. This paper introduces a method to implement variable admittance control with explicit passivity constraints using non-linear model predictive control (NMPC). The dynamics of the impedance parameters are directly included in the predictive controller. Contrary to previous approaches, the computed control signals enforce simultaneously the passivity condition and the constraints, on both the robot state and the control signals.

1. Introduction

1.1. Context

Managing physical interactions is critical for many robotic tasks, whether to achieve the compliance required for the task or for safety purposes. A classical approach involves imposing the relation between the robot's motion and the interaction force with the environment. When the imposed relation represents a mass–spring–damper second-order system, the control strategy is called impedance [1] or admittance control [2], respectively, for torque-controlled and velocity-controlled robots. This dynamical relation is fully defined by a so-called impedance profile, which is a set of interaction parameters. Thereinafter, regardless of whether admittance or impedance control is used to enforce the relationship, the parameters are referred to as impedance parameters. Although most interaction controllers implement a constant interaction behavior, i.e., constant impedance parameters, many tasks require a time-varying interaction behavior, typically to vary the apparent stiffness of the interaction in reaction to changes in the environment [3,4]. A controller that imposes a time-varying impedance profile is commonly called a variable impedance controller (VIC).

Classical impedance control with constant impedance parameters leads to stable behavior. On the contrary, time-varying impedance parameters can potentially lead to instability [5], justifying the interest in studying variable impedance/admittance control. The concept of passivity, which does not require a model of the environment, is often used to ensure the stability of the physical interaction.

This paper introduces an approach to VIC passivation that relies on predictive control to plan, online, controls that meet the VIC task objectives and respect task and joint-space constraints. In order to guarantee a passive behavior, the dynamics of the impedance parameters are also integrated into the optimization problem, hence providing a unified framework for passive VIC with constraints. Although MPC has already been used in variable impedance/admittance control, integrating passivity constraints by adapting the desired impedance/admittance profiles instead of directly altering control signals is novel, to the best of our knowledge. In the following two sections, an overview of VIC passivations as well as optimization-based VIC methods are provided, which will serve as the basis to introduce our work.

1.2. Passive variable impedance control

As previously mentioned, varying impedance profiles in time may lead to unsafe behavior [5]. In such cases, VIC control signals should be altered in some way to guarantee passivity and, consequently, stability of the physical interaction. In the following, this process is referred to as VIC passivation.

There are several approaches to implement VIC. The first one consists in modifying the impedance profile so that it can safely be imposed through classical impedance control. To do so, Kronander and Billard [5] proposed a method for offline impedance planning, where the impedance profile is evaluated beforehand, modified if deemed unsafe, and reevaluated. Bednarczyk et al. [6] proposed a similar passivation method based on impedance parameters filtering that can be used online, although with limitations on the parameters admissible variations. Most notably, the implementation from [6] is limited to diagonal impedance parameters matrices, requires a constant desired

* Corresponding author.
E-mail address: homran@unistra.fr (H. Omran).

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inertia matrix, and assumes that the stiffness and damping matrices are linearly dependent on the stiffness.

The second main category of passive VIC implementations relies on methods designed to compute robot controls directly, even from potentially unsafe impedance profiles. The most commonly used approach uses the so-called passivity tanks [7,8] that has been used for the passivation of bilateral teleoperation [9,10] and variable admittance control [11]. This method tracks dissipated energy and switches to a nominal passive profile when the energy tank is depleted. Despite its versatility, tuning passivity tanks can be task-dependent and challenging [12,13]. Furthermore, the result of the optimization is the robot control signal, such that the impedance parameters are not modified explicitly and transparently.

Finally, another approach that does not fall into either of the two previous categories was proposed by Cao et al. [14]. It consists in using the MPC to calculate a term added to the nominal control signals to meet a passivity condition. However, the resulting interaction impedance is unknown, and the properties of the nominal control can be lost.

1.3. MPC-based interaction control

Many approaches have been proposed to implement impedance and admittance control [15]. MPC is a remarkably versatile and robust method in this context, as it imposes not only the interaction behavior but also constraints on this behavior (e.g., bounds on rendered stiffness) and constraints on the robot's state (e.g., pose or velocity). Previous works proposing MPC-based interaction control have used the MPC in one of two ways: either to generate the impedance profile or to enforce it.

In the first approach, MPC is used during an intermediate stage to compute the interaction parameters, typically the rendered stiffness and damping, relevant to the desired closed-loop dynamic behavior. For instance, Haninger et al. [16] proposed an MPC framework to plan a desired trajectory and impedance parameters online from a model of uncertainties impacting a nominal plan and an estimation of force disturbance. Similarly, an MPC is used to compute the impedance parameters in [17], in a way to take into account their variations and their potential destabilization effect. Jin et al. [18] proposed an MPC-based approach to compute the impedance parameters online to regulate position tracking and interaction compliance according to some trade-off strategy. Including pose-tracking errors in the state variable enabled the introduction of ad hoc task space constraints (i.e., on Cartesian pose), even if indirectly.

The second body of work related to MPC for physical interaction concerns the actual implementation of the controller rather than the computation of its reference parameters. In this context, MPC is used to enforce state and control constraints to balance performance with safety requirements. Admittance-based force control was implemented using MPC in [19,20] to manage the interaction dynamics during a path following task, but the main goal remained regulating the contact force. Similarly, MPC has been used to limit the interaction force of position controlled robots through contact force estimation [21,22]. In [23], MPC was used to implement more traditional admittance control while mitigating contact instabilities using a model of the environment. However, it was designed to impose only the apparent damping of the robot's closed-loop behavior, not the stiffness or inertia. Bednarczyk et al. [24] proposed to use MPC to implement impedance control. The authors designed an objective cost function to impose the force–motion relationship based on the desired interaction behavior. It enforced various state and control constraints while benefiting from predictive control to obtain smoother control strategies. It should be noted that neither work considered time-varying impedance profiles.

More recently, Gold et al. [25] proposed a non-linear MPC (NMPC) method for interaction control. The authors used a prediction model for the external force based on an estimated environment and joint controllers stiffness, thus improving the proposition made in [24], where the force was assumed constant. In fact, the controller in [25] was designed for compliance and force control. The cost function was not directly related to an impedance profile, and only the apparent Cartesian stiffness could be directly imposed.

1.4. Contributions of this work

Several works have used MPC to implement time-varying impedance or admittance control. In [25], the controller switches between different manipulation primitives (MP), each consisting of a different set of weights and, therefore, different impedance parameters. However, the parameters remain constant throughout each MP, and the passivity of the transition between MP is not discussed. When the MPC is used for online impedance parameters planning, the parameters are time-varying. However, passivity considerations are not directly included in the problem formulation, except in the works of [11,14] discussed above. Typically, a separate passivation layer alters the computed impedance parameters, e.g., using a passivity tank approach as in [18].

In this work, we propose an MPC-based variable admittance controller similar to those in [24,25], but capable of providing passivity guarantees. To do so, the dynamics of the impedance parameters are included in the model used by the predictive controller. The controller imposes the dynamics of both the robot and the impedance parameters, allowing the passivity condition to be directly formulated as constraints in the NMPC optimization problem. We assume a constant inertia matrix and diagonal matrices for impedance parameters, which does not significantly limit the applicability of the method.

The proposed method for passive VIC is therefore most similar to the passivity filters [6], but it differs in that it is integrated in a predictive controller while being less restrictive about the assumed properties of the desired impedance profile. The main advantage over the other approaches, including passivity tanks and the MPC-based method proposed in [14], is that here the passivity is enforced through the alteration of the modeled impedance parameter dynamics, not through the robot control. This enables both passivity and task constraints to be satisfied concurrently, rather than sequentially, which can otherwise risk violating task constraints during passivation.

The remainder of this paper is structured as follows: After a brief introduction to serial manipulator modeling and impedance control in Section 2, the theoretical framework for the proposed controller is presented in Section 3. We illustrate the advantages of our method through comprehensive simulation and real-world experiments with a serial manipulator interacting with a human partner, in Sections 4 and 5, respectively. Finally, a discussion concludes the paper.

2. Theoretical background and notations

2.1. Serial manipulator robot modeling

Let us consider an n-joint serial robot whose motion is described by the rigid body dynamics

$$H(q)\ddot{q} + C(\dot{q}, q)\dot{q} + G(q) = \tau_c - J(q)^T w \tag{1}$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ are the joint positions, velocities, and acceleration, respectively, $\tau_c \in \mathbb{R}^n$ are the controlled joint torques, $J(q) \in \mathbb{R}^{6\times n}$ is the robot geometric Jacobian matrix [26] and $w \in \mathbb{R}^6$ is the external wrench applied to the end-effector. The inertia matrix is denoted $H(q) \in \mathbb{R}^{n\times n}$ and the Coriolis/centrifugal matrix is denoted $C(q, \dot{q}) \in \mathbb{R}^{n\times n}$. Vector $G(q) \in \mathbb{R}^n$ represents the gravity terms. The pose of the end-effector is defined in the operational space as

$$\xi \in \mathbb{R}^m = \begin{bmatrix} p \\ \phi \end{bmatrix} \tag{2}$$

where $p = [p_x, p_y, p_z]^T$ is the position and ϕ is a parameterization of the orientation, typically using the Euler angles ($\phi \in \mathbb{R}^3$ and $m = 6$) or

the quaternions ($\phi \in \mathbb{R}^4$ and $m = 7$). The end-effector twist $\nu \in \mathbb{R}^6$ can be related to $\dot{\xi}$ such that

$$\nu = [\dot{p}^T \ \omega^T]^T = J(q)\,\dot{q} \tag{3}$$

$$\dot{\xi} = [\dot{p}^T \ \dot{\phi}^T]^T = T_a(q)\nu \tag{4}$$

where $\omega \in \mathbb{R}^3$ is the angular velocity of the robot end-effector, and $T_a(q) \in \mathbb{R}^{m\times 6}$ the transformation between the geometric and analytical Jacobian matrices.

2.2. Variable admittance control

Given a desired trajectory $\{\xi^d; \dot{\xi}^d; \ddot{\xi}^d\}$, the goal of variable admittance control is to impose the task-space closed-loop dynamic behavior such that

$$M^d\ddot{e} + D^d\dot{e} + K^d e = w_a \tag{5}$$

where $w_a = -T_a^{-T}(q)w$ is the projection of w into the operational space, computed using the inverse of $T_a(q)$ mapping, and $e = \xi^d - \xi \in \mathbb{R}^m$ denotes the pose tracking errors. The time-dependent positive definite matrices M^d, D^d, $K^d \in \mathbb{R}^{m\times m}$ fully define the impedance profile such that M^d is the (constant) desired inertia matrix, D^d is the desired damping matrix, and K^d is the desired stiffness matrix. The robot's motion is imposed in response to external wrench measurements, resulting in velocity or position control of the joints.

2.3. Passivity condition for VIC

Let us consider a dynamical system of state $x(t)$ and its interaction port $\{\dot{e}; w_a\}$ with the environment. Such a system is said to be passive w.r.t. the interaction port if there exists a non-negative storage function $V(t) = V(x(t))$ such that

$$V(t) - V(0) \leq \int_0^t \dot{e}^T(\tau)w_a(\tau)d\tau \tag{6}$$

If the system is passive according to Eq. (6), then its interaction with a passive environment is likewise passive, and the resulting dynamic behavior is stable [27]. A passive VIC can, therefore, be devised by designing controllers respecting the so-called passivity condition defined by Eq. (6). Many works consider different interaction ports to reduce conservatism [5,6].

3. Passive variable admittance controller

In the following, variable admittance control is obtained using NMPC, including passivity constraints. Firstly, the control objective is derived from Eq. (5), as proposed initially in [24] by the co-authors, but with an extension to the case of time-varying parameters (see Section 3.1). Secondly, the dynamics of these parameters are included in the prediction model to form an augmented optimization problem (see Section 3.2). Finally, passivity constraints are added to ensure that the dynamics of the impedance parameters lead to a passive physical interaction (see Section 3.3).

3.1. Non-linear model predictive admittance control

Let $u \in \mathbb{R}^n$ be the robot control variables chosen as the joint accelerations, i.e., $u = \ddot{q}$, then be integrated to compute a joint velocity command, as will be done in this work. The study is restricted to the case of a constant desired inertia matrix, which, although not the general case, remains the most common in practice. Let us consider diagonal matrices M^d, $K^d(t)$, and $D^d(t)$, defining the desired impedance profile. It is assumed that M^d, D^d, and K^d are readily available, for instance resulting from offline impedance planning.

The NMPC problem for variable admittance control can be formulated as

$$\min_u \int_{t_k}^{t_k+T_p} \mathcal{L}_I(x(\tau), u(\tau))\,d\tau \tag{7a}$$

$$\text{st.} \quad x(t) = \left[q^T(t) \quad \dot{q}^T(t) \quad w^T(t)\right]^T \tag{7b}$$

$$\dot{x}(t) = f(x(t), u(t)) = \left[\dot{q}^T(t) \quad u^T(t) \quad 0_{1\times 6}\right]^T \tag{7c}$$

$$u(t) \in \mathcal{U};\ x(t) \in \mathcal{X} \tag{7d}$$

$$\xi(t) \in \mathcal{P};\ \dot{\xi}(t) \in \mathcal{V};\ \ddot{\xi}(t) \in \mathcal{A} \tag{7e}$$

where $t_k = kT_s \in \mathbb{R}_*^+$ with T_s the control sampling period; $T_p \in \mathbb{R}_*^+$ is the prediction time horizon and $f(x(t), u(t))$ is the state-space model. Here, the external wrench is assumed constant over the prediction time horizon. It should be noted that this assumption is one of the main limitations of this method and the impacts on performance will be further discussed in the conclusion. However, if a model of the environment is available or estimated online, a more realistic differential model for w can be used [25].

The stage cost $\mathcal{L}_I(t)$ associated with accurately tracking the interaction behavior can be written as

$$\mathcal{L}_I(t) = \left\| M^d\ddot{e} + D^d\dot{e} + K^d e - w_a \right\|_{R_I}^2 \tag{8}$$

omitting time-dependency to simplify the notation, where $\|a\|_{R_I}^2$ denotes the weighted norm $a^T R_I a$, with $R_I \in \mathbb{R}^{m\times m}$ a positive definite gain matrix.

The control and state variables are constrained within the convex sets $\mathcal{U}$ and $\mathcal{X}$, respectively. Likewise, $\mathcal{P}$, $\mathcal{V}$, and $\mathcal{A}$ are sets used to constrain the robot Cartesian pose, velocity, and acceleration, respectively, within admissible regions. An additional term $\mathcal{L}_\varnothing(t)$ can be added to the stage cost to fulfill, when applicable, a secondary objective (e.g., elbow position of a redundant robot).

It should be noted that the state-space model $f(x(t), u(t))$, at the exception of the derivative of the wrench $w(t)$, is expressed in joint space. Nonetheless, the optimization problem is mostly performed in cartesian space, the robot kinematic model is used to evaluate the cost function and constraints. For instance, the relation between $x(t)$ and $\dot{\xi}(t)$ is defined by Eqs. (3) and (4).

3.2. Impedance parameters dynamics

The naive NMPC formulation (7) of VIC applies the desired impedance parameters regardless of the passivity of the resulting interaction behavior. The desired parameters should therefore be modified so as to meet the passivity condition. To that end, Eq. (5) is modified to use filtered parameters M, $K(t)$, and $D(t)$ instead of the potentially unsafe desired ones such that

$$M\ddot{e}(t) + D(t)\dot{e}(t) + K(t)e(t) = w_a(t) \tag{9}$$

The process of VIC passivation consists then in computing these modified parameters to track the desired ones, while guaranteeing that the interaction remains passive.

The desired and current impedance parameters are the diagonal terms of the respective matrices. For instance, $K^d(t)$ and $K(t)$ matrices are parameterized as

$$K^d(t) = \text{diag}\{k_1^d(t),\ k_2^d(t),\ \ldots,\ k_m^d(t)\}$$

$$K(t) = \text{diag}\{k_1(t),\ k_2(t),\ \ldots,\ k_m(t)\}$$

where $\text{diag}\{\cdot\}$ denotes the function that maps the vector of diagonal terms to the diagonal matrix. The matrices $D^d(t)$ and $D(t)$ are parameterized in the same way. The constant inertia terms are chosen such that $M = M^d$.

For the impedance parameters to track the desired parameters, a first-order tracking law is imposed for the terms of $K(t)$ and $D(t)$ such that

$$\dot{k}_i(t) = \beta(t)\left(k_i^d(t) - k_i(t)\right),\ i \in \{1, \dots, m\} \tag{10}$$

$$\dot{d}_i(t) = \beta(t)\left(d_i^d(t) - d_i(t)\right),\ i \in \{1, \dots, m\} \tag{11}$$

where $\beta(t) \in \mathbb{R} \geq 0$ is a time-varying gain. With this formulation, the gain β effectively controls the dynamics of the impedance parameters: for $\beta = 0$, the parameters will remain constant and the desired values will be ignored ; for $\beta \geq 0$, the impedance parameters will track the desired ones. The larger the value of β, the better the tracking becomes, i.e., reduced delay between desired and effective impedance parameters, such that the maximization of β becomes a secondary objective of the NMPC problem in (12) below.

The optimization problem (7) is augmented with the dynamics of the impedance parameters, such that:

$$\min_{\overline{u}} \int_{t_k}^{t_k+T_p} \left(\mathcal{L}_I\left(\overline{x}(\tau), \overline{u}(\tau)\right) + \mathcal{L}_\beta\left(\overline{u}(\tau)\right)\right) d\tau \tag{12a}$$

$$\text{st.} \quad \overline{x}(t) = \begin{bmatrix} x^T(t) & k^T(t) & d^T(t) \end{bmatrix}^T \tag{12b}$$

$$\dot{\overline{x}}(t) = \begin{bmatrix} f\left(x(t), u(t)\right) \\ \beta(t)\left(k^d(t) - k(t)\right) \\ \beta(t)\left(d^d(t) - d(t)\right) \end{bmatrix} \tag{12c}$$

$$\overline{u}(t) = \begin{bmatrix} u^T(t) & \beta(t) \end{bmatrix} \tag{12d}$$

$$\overline{x}(t) \in \overline{\mathcal{X}} \tag{12e}$$

$$u(t) \in \mathcal{U};\ \beta(t) \in \left[0, \beta_{max}\right] \tag{12f}$$

$$\xi(t) \in \mathcal{P};\ \dot{\xi}(t) \in \mathcal{V};\ \ddot{\xi}(t) \in \mathcal{A} \tag{12g}$$

$$h(t) = h(\overline{x}(t), \overline{u}(t)) \geq 0 \tag{12h}$$

where $\overline{x}$ and $\overline{u}$ are the state and control variables, augmented with the impedance parameters and time-varying gain $\beta(t)$, respectively. As a reminder, solving the optimization problem provides the control inputs $\overline{u}$ over the prediction horizon, which correspond to the regulated joint accelerations per (7) and the gain of the low-pass filter applied to the impedance parameters. The variables $k(t)$ and $d(t)$ are the vectors of stiffness parameters $k_i(t)$ and damping parameters $d_i(t)$, respectively. The cost associated with tracking the desired parameters is

$$\mathcal{L}_\beta\left(\overline{u}\right) = \left\|\beta_{max} - \beta(t)\right\|_{R_\beta}^2 \tag{13}$$

where $R_\beta \in \mathbb{R}_*^+$ is a constant weight. This maximization of β, up to β_{max}, besides ensuring numerical stability, results in a maximized impedance parameters tracking performance. Eq. (12h) defines an additional constraint $h(t) = h(\overline{x}(t), \overline{u}(t)) \geq 0$ where $h(t) \in \mathbb{R}$ is a scalar indicator for the passivity of the interaction. This constraint is used in the following section to enforce passivity by reducing the $\beta(t)$ when necessary.

3.3. Impedance profile passivation

Function $h(t)$ will now be devised to guarantee the passivity of the interaction, such that $h(t) \geq 0$ is a sufficient condition for Eq. (6). A passivity condition is defined by analogy with the work of Kronander and Billard [5] that considers the passivity of the interaction port $\{r; w_a\}$, where $r(t)$ is a filtered error variable $r = \dot{e} + \alpha e$, with $\alpha \in \mathbb{R}^+$ constant. Similarly, we consider the positive definite continuously differentiable storage function defined as

$$V(t) = \frac{1}{2} r^T(t) M r(t) + \frac{1}{2} e^T(t) P(t) e(t) \tag{14}$$

where

$$P(t) = K(t) + \alpha D(t) - \alpha^2 M \tag{15}$$

It yields

$$\int_0^t r^T(\tau) w_a(\tau) d\tau = V(t) - V(0) + \int_0^t h(\tau) d\tau \tag{16}$$

using Eq. (9) and the definition of r, with

$$h(t) = \dot{e}^T(t)\left(D(t) - \alpha M\right)\dot{e}(t) + e^T(t)\left(\alpha K(t) - \frac{1}{2}\dot{P}(t)\right)e(t) \tag{17}$$

From (16) and the inequality $h(t) \geq 0$ enforced by the MPC in Eq. (12h), we obtain the passivity of the controlled robotic system from input w_a to output r :

$$V(t) - V(0) \leq \int_0^t r^T(\tau) w_a(\tau) d\tau \tag{18}$$

In Eq. (16), the left term represents the energy that is injected into the system by the input–output interaction port $\{w_a, r\}$. A part of this energy is stored in the storage function $V(\cdot)$. The other part, represented by the integral of the $h(\cdot)$, represents the dissipated energy. Note that Eq (16) implies

$$\dot{V}(t) = r^T(t) w_a(t) - h(t) \leq r^T(t) w_a(t) \tag{19}$$

The positivity of the quantity $h(\cdot)$, which represents the dissipative power, implies that the rate of change of the storage function is not higher than the supply rate from the port. The imposed passivity property guarantees that no internal energy is generated due to the time variations of the impedance profiles. In case where $w_a = 0$, the storage function satisfies $\dot{V}(t) \leq 0$. By considering V as a candidate Lyapunov function, the stability of the equilibrium point follows from Theorem 4.8 in [28]. This approach, analogous to the proof of Theorem 1 in [5], demonstrates that the system remains stable under time-varying impedance profiles during free motion.

To ensure the existence of a solution, α is chosen such that $D(t) - \alpha M \geq 0,\ \forall t$. To that end, we choose α similarly to [5] such that

$$\alpha = \frac{\min(\lambda_{D^d})}{\max(\lambda_M)} \tag{20}$$

where $\min(\lambda_{D^d})$ and $\max(\lambda_M)$ are the smallest eigenvalue of $D^d(t)\,\forall t$ and the largest eigenvalue of M, respectively.

3.4. Practical implementation

The optimization problem of Eq. (12) can be solved using off-the-shelf non-linear NMPC solvers. In the following, the solver acados [29] is used to solve the optimization with a direct multiple shooting approach and HPIPM [30] as the underlying QP solver. The shooting nodes are uniformly sampled over the prediction horizon such that

$$T_p = N T_s \text{ (s)} \tag{21}$$

where $N \in \mathbb{N}^*$ is the number of shooting nodes.

4. Illustrative simulation results

The method is first evaluated in simulation, benefiting from fully reproducible interactions necessary to compare different scenarios. A lower-dimensional space (2D) is chosen to simplify the analysis and visualization of the controller behavior.

4.1. Simulation scenario

As represented in Fig. 1, a planar robot with two revolute joints is simulated such that $n = m = 2$, $\xi = [p_x, p_y]^T$, and $T_a(q) = I_{2\times2}$. This robot has two identical links of unit length. The proposed NMPC-based variable admittance controller is implemented as detailed in Section 3 using the acados solver. The controller parameters are set to $T_s = 5$ ms (200 Hz update rate), $T_p = 50$ ms (i.e., $N = 10$), $R_I = 10^3 I_{2\times2}$, $R_\beta = 1$, and $\beta_{max} = 100$. Constraints are set for the joint acceleration and the Cartesian pose such that

$$\ddot{q}_i(t) \in [-3,\ 3]\ (\text{rad.s}^{-2}),\ i \in \{1;\ 2\} \tag{22}$$

$$p_x(t) \in [0.9,\ 2]\ (\text{m}) \tag{23}$$

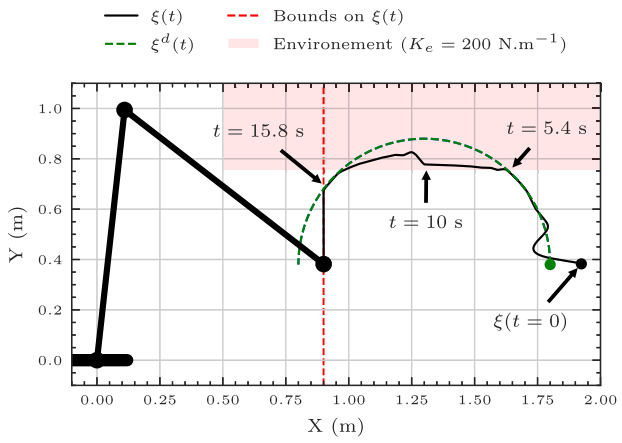


Fig. 1. Simulated Cartesian robot trajectory $\xi(t)$ is represented with a solid black line and desired trajectory $\xi^d(t)$ with the dotted green line. The robot, environment, relevant constraints, and time stamps are also represented. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$p_y(t) \in [0,\ 1]\ \text{(m)} \tag{24}$$

The robot interacts with an elastic environment of stiffness $K_e = 200$ N.m^{-1}, which is simulated such that the external force is $w(t) = \begin{bmatrix}0 & w_y(t)\end{bmatrix}^T$ (N), with

$$w_y(t) = \begin{cases} K_e(0.75 - p_y(t)) & \text{if } p_y(t) \geq 0.75 \\ 0 & \text{otherwise} \end{cases} \tag{25}$$

The desired trajectory follows a half-circle in the task space such as reported in Fig. 1. The desired impedance profile is defined as

$$M = \text{diag}\{5,\ 5\}\ \text{(kg)} \tag{26}$$

$$K^d(t) = \text{diag}\{50,\ k_2^d(t)\}(\text{N.m}^{-1}) \tag{27}$$

$$D^d(t) = \text{diag}\{d_1^d(t),\ d_2^d(t)\}(\text{N.m}^{-1}\text{.s}) \tag{28}$$

where

$$k_2^d(t) = \begin{cases} 50, & \text{if } t \leq 10\ \text{s} \\ 250, & \text{otherwise} \end{cases} \tag{29}$$

The damping terms are computed from M and $K^d(t)$ to maintain a damping ratio $\zeta = 0.2$ such that

$$d_i^d(t) = 2\zeta\sqrt{m_i k_i^d(t)},\ i \in \{1,\ 2\} \tag{30}$$

It should be noted that this low damping ratio is chosen to highlight passivity issues and make the effect of passivation more apparent on the plots.

4.2. Results

The tracking errors reported in Fig. 2 indicate that when no external force is applied, the robot converges towards the desired pose with the expected closed-loop behavior. Additionally, the commanded joint accelerations $\ddot{q}_2(t)$ are correctly bounded, as can be seen at startup in Fig. 3. The interaction with the elastic environment starts at $t = 5.4$ s (see Fig. 1), resulting in an increasing external force and pose tracking error due to the relatively low interaction stiffness (see Fig. 2). However, until the change in desired impedance parameters at $t = 10$ s, the interaction behavior remains passive with no need for corrective actions, as indicated by the positiveness of $h(t)$ in Fig. 4.

At $t = 10$ s, the desired stiffness is increased (see Fig. 4, center), and so does the desired damping as per Eq. (30). Without any particular action, such change in the desired impedance profile would lead to a loss of passivity. With the proposed method, the dynamics of the impedance parameters are altered through a reduced value of $\beta(t)$ to maintain the positiveness of $h(t)$ (see Fig. 4). The impedance parameters then converge towards their desired values in 0.5 s, while the constraint on $h(t)$ maintains a passive interaction.

After $t = 10.5$ s, the desired interaction behavior once again requires no corrective actions to remain passive, as the desired stiffness is constant. Around 15.8 s, the end-effector reaches the Cartesian constraint

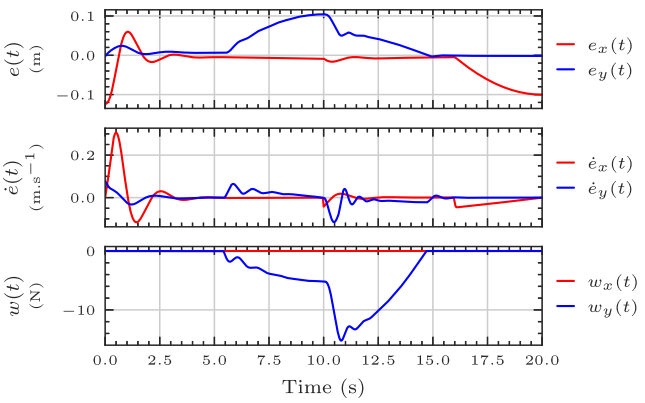


Fig. 2. Top and center: position and velocity tracking errors, respectively denoted $e(t)$ and $\dot{e}(t)$. Bottom: simulated external force measurements $w(t)$.

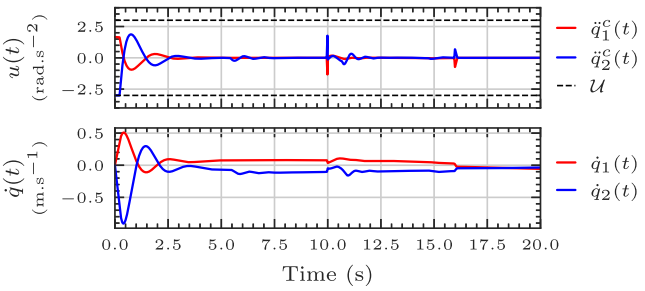


Fig. 3. Top: NMPC control variable $u(t) = \ddot{q}(t)$. Bottom: resulting (commanded) robot joint velocities $\dot{q}(t)$.

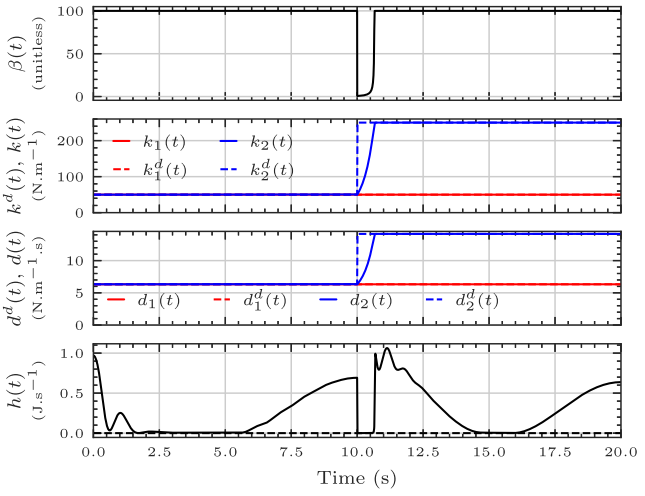


Fig. 4. Top: control variable $\beta(t)$. Center: desired and altered impedance parameters. Only the stiffness and damping are reported, because inertia is constant. Bottom: passivity condition $h(t)$ used in the passivity constraint $h(t) \geq 0$.

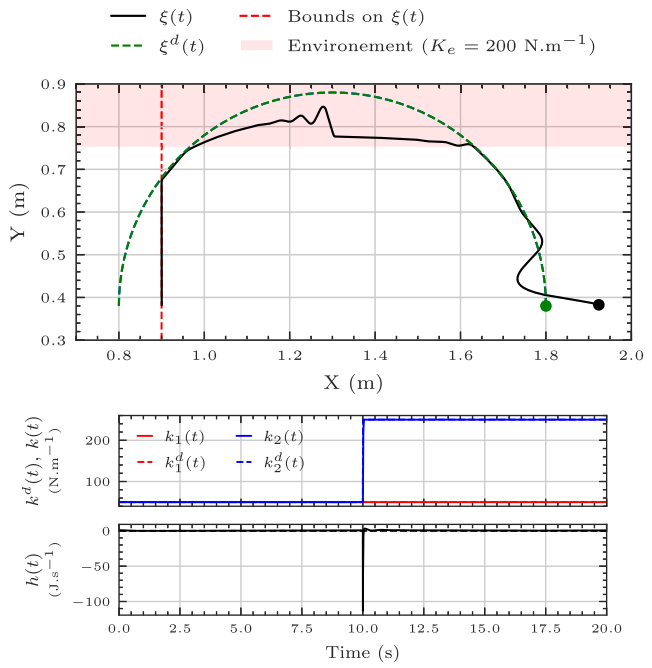


Fig. 5. Top: task-space behavior if the passivity constraint is not enforced. Center and bottom: associated stiffness parameters and passivity condition.

imposed in $p_x \geq 0.9$. As visible in Fig. 1, system control then results in a trajectory respecting this constraint.

4.3. Comparison with non-passive NMPC

The previous simulation is replicated without the constraint (12h) so that the passivity condition is not considered. When the impedance parameters are constant (i.e., at all times except around $t = 10$ s), the interaction behavior is identical to that of its passive counterpart (see Fig. 5, top). However, at $t = 10$ s, the change in desired stiffness (and damping) is implemented without any filtering, resulting in a highly non-passive behavior as revealed by the value reached by $h(t)$ (see Fig. 5, bottom). In this particular scenario, the loss of passivity does not cause an unstable interaction, even though the transient trajectory is more oscillatory (see Fig. 5, top). This can be explained by the globally dissipative behavior that results from the damping dissipating over time the energy introduced by the increased stiffness. Nonetheless, stability cannot be guaranteed in the case of such a passivity violation [5].

4.4. Discussion

The simulation results demonstrate the applicability of the proposed approach to implement variable admittance control while guaranteeing passivity and allowing the definition of state and control constraints. Another advantage of this method is that, due to the low-pass filter formulation of their dynamics, the impedance parameter values are guaranteed to be continuous even if $\beta(t)$ is potentially discontinuous (as is the case at $t = 10$ s in Fig. 4). Furthermore, the time derivative of the impedance parameters can be bounded, either indirectly, using β_{max} as per Eq. (12), or directly, using an explicit constraint added to the NMPC. The augmented state constraints can likewise be devised to bound the impedance parameters themselves, for instance to guarantee a minimal stiffness or damping as in [31].

It is important to comment on the main difference and the expected behavior between the proposed approach and the widely used passivity tank methods. A passivity tank stores the energy dissipated by the system, up to a threshold, and uses it to perform non-passive action, e.g., increasing the stiffness. As long as the tank is not depleted, it allows any parameter variations. Therefore, it is usually possible to choose passivity tank parameters, i.e., initial stored energy and storage threshold, so that the impedance parameters are applied transparently and independently of instantaneous passivity considerations. This can lead to a passive, but practically unstable interaction if the parameters are poorly chosen [32]. On the other hand, the controller stops performing the action once the tank is depleted, which can lead to discontinuous controls. For instance, in [7], the stiffness matrix is defined as the sum of constant and variable terms such that $K(t) = K_0 + K_v(t)$; whenever the tank is depleted, $K_v(t)$ is set to zero. With our approach, there is no task-specific tuning and the passivation of a sudden change in desired stiffness still results in smooth controls.

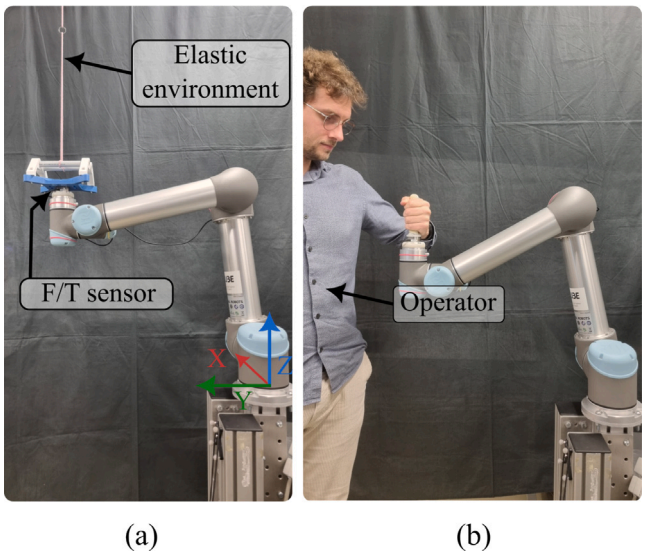


Fig. 6. Experimental setups.

5. Experimental results

The performance of the proposed controller is now evaluated on a robotic platform by varying the desired impedance profile during an interaction with an elastic environment. A second experiment is then conducted with a human operator to demonstrate the applicability to physical human–robot interaction (pHRI) scenarios.

5.1. Description of the robotic setup and the experiments

The passive variable admittance controller is implemented using a UR5 Universal Robot 6-DOF manipulator ($n = 6$). The end-effector of the robot is equipped with a force torque sensor (ATI mini 40). During the first experiment, a 3D printed part is mounted onto the robot end-effector to attach an elastic cord whose other end is attached to the ceiling (Fig. 6(a)). During the second experiment, a human operator manipulates a handle attached to the robot end-effector (Fig. 6(b)).

Similarly to Section 4, the acados solver is used to model the optimization problem and subsequently generate code imported in a C++ controller based on the *ros-control* framework. This controller runs on a laptop with a 2.30 GHz Intel Core i9-12950HX processor. The controller sampling time is chosen as $T_s = 8$ ms to match the 125 Hz rate of the UR5 driver. The NMPC is implemented with $T_p = 80$ ms ($N = 10$), $R_I = 10^5 I_{6\times6}$, $R_\beta = 1$, and $\beta_{max} = 100$. Control and state constraints are set as

$$u(t) \in [-3, 3] \text{ (rad.s}^{-2}\text{)}, \; i \in \{1, 2, \ldots, 6\} \tag{31}$$

$$p_z(t) \in [0.45, \; 1] \text{ (m)} \tag{32}$$

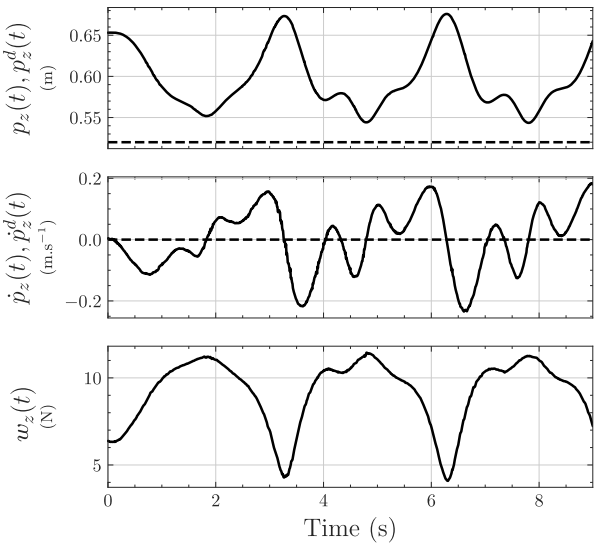


Fig. 7. Robot pose, velocity, and external wrench during experiment (a).

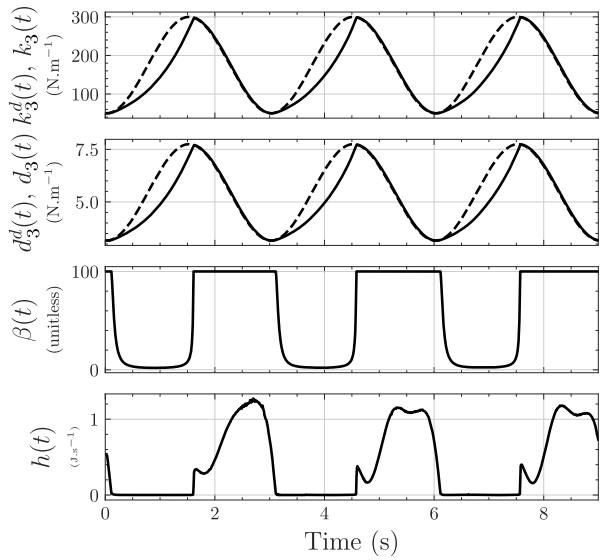


Fig. 8. Impedance profile (only stiffness and damping reported) and passivity constraints-related variables during experiment (a).

The end-effector orientation is parameterized with Euler yaw-pitch-roll angles, i.e., $\phi \in \mathbb{R}^3$ ($n = m = 6$). The desired pose is kept constant, i.e., $\ddot{\xi}^d(t) = \dot{\xi}^d(t) = 0$ and the desired impedance profile is defined as

$$M = \text{diag}\{5,\ 5,\ 5,\ 0.5,\ 0.5,\ 0.5\}\ (\text{kg/kg.m}^2) \tag{33}$$

$$K^d(t) = \text{diag}\{50,\ 50,\ k_3^d(t),\ 5,\ 5,\ 5\}\ (\text{N.m}^{-1}) \tag{34}$$

where $k_3^d(t)$ is the time-varying stiffness along the z-axis such that $k_3^d(t) = 50\,(1 - s(t)) + 300\,s(t)$ with

$$s(t) = 0.5\left(1 - \cos\frac{2\pi}{3}t\right) \tag{35}$$

The desired damping coefficients $d_i^d(t)$ are computed to maintain a low damping ratio $\zeta = 0.1$ chosen to highlight passivity issues without the need for high-frequency or amplitude stiffness variations.

5.2. Experiment (a): interaction with an elastic environment

At startup, the interaction stiffness is at its lowest (i.e., 50 N.m^{-1}) such that, due to the force applied by the elastic cord, the pose tracking error is at its highest (see Fig. 7).

When the desired stiffness and damping along the z-axis increase (see Fig. 8), the impedance parameters accurately track the desired value until the passivity constraint is no longer respected at $t = 0.2$ s. Once $h(t)$ reaches zero, the impedance profile is altered to maintain the positiveness of $h(t)$ as visible in Fig. 8 for $t \in [0.2, 1.5]$ s.

When the desired stiffness decreases, e.g., for $t \in [1.5, 3]$ s, the interaction behavior is dissipative due to the damping and motion, as evidenced by the strictly positive corresponding value of $h(t)$. As a result, the value of $\beta(t)$ is maximized, and the impedance profile is tracked accurately. The closed-loop dynamic behaviors of the robot, as well as the evolution of the impedance parameters imposed by the controller are consistent and repeatable across the three periods reported for this experiment.

The control loop, NMPC computation included, takes 5.6 ms on average, which is under the 8 ms required as per the sampling time, but close to maximum that could be achieved on this hardware (see Table 1).

5.3. Experiment (b): pHRI

Now consider the interaction with a human operator corresponding to the setup (b) in Fig. 6. The controller, desired trajectory, and impedance profile remain the same as in the previous experiment. During the first period of impedance variation (i.e., $t \in [0, 3]$ s), the operator does not interact with the robot. Contrarily to methods that are not based on the actual state of the system [5,6], the proposed approach allows impedance variations when the tracking errors are low or null, as $h(t)$ will likewise be small or null. This is visible in Fig. 9 where $h(t)$ is null and the value of $\beta(t)$ is maximized such that the desired impedance profile is accurately tracked.

During the second period of impedance variation (i.e., $t \in [3, 6]$ s), the operator applies a force along the z-axis to move the end-effector upward. As pose tracking errors appear, the desired stiffness variation would result in a non-passive interaction behavior. As a result of the

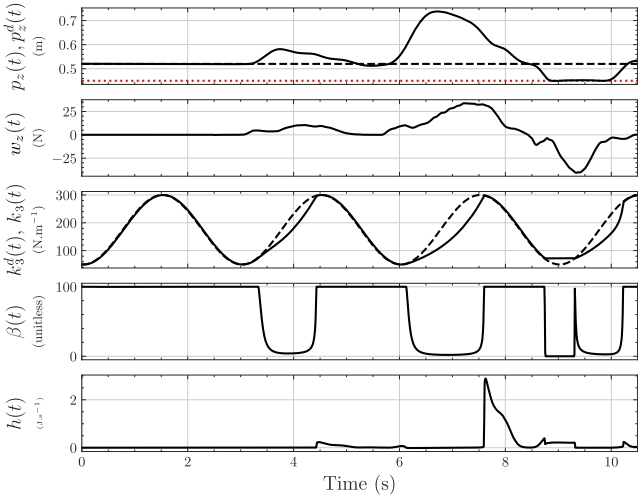


Fig. 9. Data collected during the pHRI experiment (b). The red dotted line refers to a constraint. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
NMPC computation time for different horizon lengths.

N	1	2	3	4	5	10	15
avg (ms)	0.7	1.3	1.8	2.4	2.9	**5.6**	9
std (ms)	0.04	0.04	0.1	0.1	0.1	**0.4**	0.5

proposed control, it is altered through a lower value of $\beta(t)$ to maintain the positiveness of $h(t)$. The interaction is then replicated with a larger force (i.e., $t \in [6, 8]$ s), leading to a similar result. Finally, a negative force is applied along the z-axis during the last period (at $t = 8.5$ s). Due to the Cartesian task-space constraint on $p_z(t)$, the motion of the end-effector is halted at $p_z(t) = 0.45$ m. Additionally, the impedance parameters are altered to maintain the positiveness of $h(t)$. When the operator stops interacting with the robot (around $t = 10$ s), the end-effector returns to its desired position according to the desired closed-loop dynamics.

It can be noted that there is an interaction between the task space constraints (e.g., position) and the impedance parameter dynamics, visible, for instance, at $t = 8.75$ s in Fig. 9, when the tracking of the desired parameters is halted as the position limit is reached. This is because a decrease in stiffness would result in a larger predicted position tracking error, hence violating the constraint.

6. Conclusions and discussion

In this paper, we proposed an NMPC formulation for the variable admittance control of robot manipulators. The dynamics of the variable impedance parameters (i.e., stiffness and damping terms) were included in the problem formulation to track the desired parameters while guaranteeing a passive interaction. This passivity condition is imposed through a constraint in the NMPC, allowing the online adaptation of the planned variable impedance parameters for passivity. The approach was shown to be adapted for variable admittance control with state and control variable constraints, including task space constraints such as bounds on Cartesian pose and velocity. This makes the method especially well suited for pHRI, as it allows to meet both passivity and safety-related task constraints (e.g., bounds on robot velocity or acceleration) in a unified framework, all the while guaranteeing smooth impedance parameter variations.

The applicability of the proposed approach has been demonstrated by simulation. The variable admittance control has been implemented while respecting a set of state and control constraints and simultaneously guaranteeing the passivity of the interaction. The controller was also successfully implemented with a widespread collaborative robotic platform and the experimental results assessed the applicability of the approach in practical scenarios, including the interaction with an operator. It is important to emphasize that the proposed approach for passive admittance control is generic. The controller could readily be extended to impedance control of torque-controlled robots by directly using the Euler–Lagrange model defined by Eq. (1) and choosing the torques τ_c as the control variable.

With the proposed implementation, a standing limitation lies in the modeling of the external wrench, as the wrench was considered a constant disturbance over the prediction horizon in Eq. (7c). As such, it is implicitly assumed that the wrench does not vary too much over the prediction horizon, hence limiting applications to human–robot physical interactions and low stiffness environments in general. In addition, it is not possible to constrain the force applied to the environment as it is considered constant over the prediction horizon. It should be noted that a differential model for the external wrench (e.g., through environment stiffness estimation [25,33]) could be integrated to the proposed approach without any significant change besides the model of $w(t)$ in Eq. (7c). However, including the model of the wrench would require the estimation of a model of interaction between the robot and its environment, which remains a largely open problem for complex interactions.

It is also worth mentioning that the proposed controller based on NMPC requires the desired impedance profile to be known at least over the prediction horizon. If the profile is planned online from sensor data, then only the current desired values might be available. In that case, the desired parameters would be constant over the horizon and it might be advantageous to separate the control of parameters dynamics from the control of the robot motion in order to reduce computation time. This will be investigated by the authors in future works.

CRediT authorship contribution statement

Thibault Poignonec: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Conceptualization. **Bernard Bayle:** Writing – review & editing. **Loïc Cuvillon:** Writing – review & editing. **Hassan Omran:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hassan Omran reports financial support was provided by French National Research Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.robot.2026.105483.

Data availability

Data will be made available on request.

References

[1] N. Hogan, Impedance control: An approach to manipulation, J. Dyn. Syst. Meas. Control. 107 (1) (1985) 1–24.
[2] W.S. Newman, Stability and performance limits of interaction controllers, J. Dyn. Syst. Meas. Control. 114 (4) (1992) 563–570.
[3] M. Sharifi, A. Zakerimanesh, J.K. Mehr, A. Torabi, V.K. Mushahwar, M. Tavakoli, Impedance variation and learning strategies in human–robot interaction, IEEE Trans. Cybern. 52 (7) (2022) 6462–6475.
[4] M.J. Pollayil, F. Angelini, G. Xin, M. Mistry, S. Vijayakumar, A. Bicchi, M. Garabini, Choosing stiffness and damping for optimal impedance planning, IEEE Trans. Robot. 39 (2) (2023) 1281–1300.
[5] K. Kronander, A. Billard, Stability considerations for variable impedance control, IEEE Trans. Robot. 32 (5) (2016) 1298–1305.
[6] M. Bednarczyk, H. Omran, B. Bayle, Passivity filter for variable impedance control, in: International Conference on Intelligent Robots and Systems, IROS, IEEE, 2020, pp. 7159–7164.
[7] F. Ferraguti, N. Preda, A. Manurung, M. Bonfè, O. Lambercy, R. Gassert, R. Muradore, P. Fiorini, C. Secchi, An energy tank-based interactive control architecture for autonomous and teleoperated robotic surgery, IEEE Trans. Robot. 31 (5) (2015) 1073–1088.
[8] Z. Jin, A. Liu, W.-a. Zhang, L. Yu, An optimal variable impedance control with consideration of the stability, IEEE Robot. Autom. Lett. 7 (2) (2022) 1737–1744.
[9] N. Piccinelli, R. Muradore, A bilateral teleoperation with interaction force constraint in unknown environment using non linear model predictive control, Eur. J. Control 62 (2021) 185–191.
[10] N. Piccinelli, R. Muradore, Linearized virtual energy tank for passivity-based bilateral teleoperation using linear MPC, IEEE Trans. Robot. 41 (2025) 2589–2604.
[11] C. Secchi, F. Ferraguti, Energy optimization for a robust and flexible interaction control, in: International Conference on Robotics and Automation, ICRA, IEEE, 2019, pp. 1919–1925.
[12] F. Califano, R. Rashad, C. Secchi, S. Stramigioli, On the use of energy tanks for robotic systems, Int. Work. Human-Friendly Robot., HFR (2022) 174–188.
[13] R. Zanella, G. Palli, S. Stramigioli, F. Califano, Learning passive policies with virtual energy tanks in robotics, IET Control Theory Appl. 18 (5) (2024) 541–550.

[14] R. Cao, L. Cheng, H. Li, Passive model-predictive impedance control for safe physical human–robot interaction, IEEE Trans. Cogn. Dev. Syst. 16 (2) (2024) 426–435.
[15] M. Schumacher, J. Wojtusch, P. Beckerle, O. Von Stryk, An introductory review of active compliant control, Robot. Auton. Syst. 119 (2019) 185–200.
[16] K. Haninger, C. Hegeler, L. Peternel, Model predictive impedance control with Gaussian processes for human and environment interaction, Robot. Auton. Syst. 165 (2023) 104431.
[17] N. Thelenberg, C. Ott, On handling variable stiffness parameters in compliance control via MPC, in: European Control Conference, ECC, IEEE, 2024, pp. 615–620.
[18] Z. Jin, D. Qin, A. Liu, W.-A. Zhang, L. Yu, Model predictive variable impedance control of manipulators for adaptive precision-compliance tradeoff, IEEE/ASME Trans. Mechatronics 28 (2) (2023) 1174–1186.
[19] K.J. Kazim, J. Bethge, J. Matschek, R. Findeisen, Combined predictive path following and admittance control, in: American Control Conference, ACC, IEEE, 2018, pp. 3153–3158.
[20] J. Matschek, J. Bethge, P. Zometa, R. Findeisen, Force feedback and path following using predictive control: Concept and application to a lightweight robot, IFAC-PapersOnLine 50 (1) (2017) 9827–9832.
[21] N. Piccinelli, R. Muradore, Interaction force constraints for position-controlled manipulator using linear MPC, in: 21st International Conference on Advanced Robotics, ICAR, IEEE, 2023, pp. 176–182.
[22] S. Kleff, E. Dantec, G. Saurel, N. Mansard, L. Righetti, Introducing force feedback in model predictive control, in: International Conference on Intelligent Robots and Systems, IROS, IEEE, 2022, pp. 13379–13385.
[23] A. Wahrburg, K. Listmann, MPC-based admittance control for robotic manipulators, in: 55th Conference on Decision and Control, CDC, IEEE, 2016, pp. 7548–7554.
[24] M. Bednarczyk, H. Omran, B. Bayle, Model predictive impedance control, in: International Conference on Robotics and Automation, ICRA, IEEE, 2020, pp. 4702–4708.
[25] T. Gold, A. Völz, K. Graichen, Model predictive interaction control for robotic manipulation tasks, IEEE Trans. Robot. 39 (1) (2023) 76–89.
[26] B. Siciliano, O. Khatib, Springer Handbook of Robotics, Springer, 2016.
[27] R. Sepulchre, M. Jankovic, P.V. Kokotovic, Constructive Nonlinear Control, Springer Science & Business Media, 2012.
[28] H.K. Khalil, Nonlinear Systems, third ed., Prentice Hall, 2002.
[29] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N.v. Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, M. Diehl, Acados — a modular open-source framework for fast embedded optimal control, Math. Program. Comput. 14 (1) (2022) 147–183.
[30] G. Frison, M. Diehl, HPIPM: a high-performance quadratic programming framework for model predictive control, IFAC-Pap. 53 (2) (2020) 6563–6569.
[31] Z. Jin, D. Qin, A. Liu, W.-A. Zhang, L. Yu, Constrained variable impedance control using quadratic programming, in: International Conference on Robotics and Automation, ICRA, IEEE, 2022, pp. 8319–8324.
[32] F. Benzi, F. Ferraguti, G. Riggio, C. Secchi, An energy-based control architecture for shared autonomy, IEEE Trans. Robot. 38 (6) (2022) 3917–3935.
[33] M.D. Killpack, A. Kapusta, C.C. Kemp, Model predictive control for fast reaching in clutter, Auton. Robots 40 (3) (2016) 537–560.